



SVKM'S NMIMS

**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT**

Academic Year: 2023-2024

Program/s: BTech/MBA Tech

Year: II Semester: IV

Stream/s ; Data Science/CSE DS _____

Subject: Introduction to Data, Signal and Image Analysis

Time: 03 hrs (10 am to 1 pm)

Date: 3/7/24

No. of Pages: 03

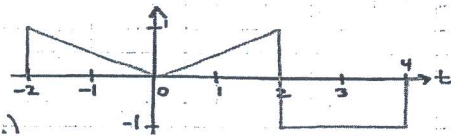
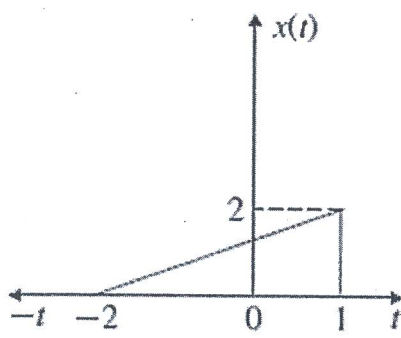
Marks: 100

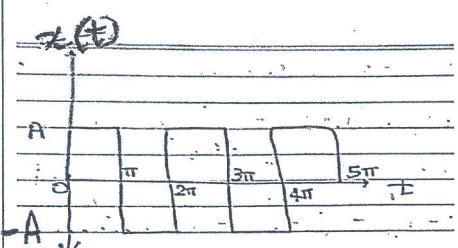
Re-exam 2022-23

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1, BL-3, SO1	a.	Apply Z transform on signal:- $n.u[n]$	[5]
CO-1, BL - 3, SO2	b.	Solve to find $x(0)$ and $x(\infty)$ for :- $X(S) = \frac{2S+6}{s(3S+4)}$	[5]
CO2, BL1, SO1	c.	List types of digital filters explain how LTI system works as a filter	[5]
Co1, BL1, SO1	d.	Write a note on Analog and Digital signals.	[5]
Q2 CO-1, BL-1; CO-3, BL-3, SO1	A	What do you understand by Causal, Anti-causal? Give example on each Apply sampling concept on a continuous time signal $x(t)=5\sin 200\pi t$ and find W_m, f_m, f_s .	[10]

Q2 CO1,BL3 ,SO1	B	 Consider the signal $x(t)$ given in the figure having value of +1 and -1 on the y-axis and x axis range from -2 to +4. Perform and sketch $x(-t)$ and $x(2t+3)$	[10]
Q3 CO-3, BL-5,SO2	A	Determine $Y(t)$ by using integral convolution. Given – $x(t) = t.u(t)$ and $h(t) = e^{-5t}u(t)$	[10]
Q3 CO-1, BL-5,SO2	B	Evaluate- $L[\int_0^t t.e^{3t} \cos 5t dt]$	[10]
Q4 CO1,BL3 ,SO1	A	 Construct and sketch the signal $5x(-3t+1)$ where $x(t)$ is shown in the above figure.	[10]
Q4 CO-1, BL-1,SO2	B	If $X(S) = \frac{1}{(s^2 - 3s + 2)}$ Calculate and find $x(t)$ for all possible ROC.	[10]
Q5 CO-2, BL-2,SO1	A	What are the advantages and disadvantages of impulse invariant transformation? Explain ideal characteristics of filter.	[10]
Q5 CO4,BL4 ,SO1	B	What is bit plane slicing? Explain bit plane slicing for the given image:- $\begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 1 & 2 & 4 \\ 5 & 1 & 6 & 2 \\ 2 & 3 & 5 & 6 \end{bmatrix}$	[10]

Q6 CO-2, BL-4,SO1	A	Differentiate FIR filter vs IIR filter	[10]
Q6 CO4, BL5,SO1	B	"Image enhancement is significant and is widely used in numerous applications". Support the above statement with valid explanation.	[10]
Q7 CO4,BL4, SO1	A	Write a note on JPEG and PNG image formats. Compare Rank vs Gaussian filter.	[10]
Q7 CO-1, BL- 1,BL3,SO1	B	Write 3 applications of Fourier Transform. Apply Fourier series(trigonometric form) for the given periodic signal. The value of the signal is +A and -A on the positive and negative (y) axis respectively. The points on x(t-axis) are $\pi, 2\pi, 3\pi, 4\pi, 5\pi..$ 	[10]